WATER QUALITY AND OXYGEN 20

The quality of water is determined by many factors. Two important considerations are the amount of oxygen dissolved in the water sample and the **biological oxygen demand** (BOD) of that sample. BOD is an indication of how much oxygen is needed by organisms in a sample of water.

Oxygen becomes dissolved in water in many ways; including by diffusing from the atmosphere, by aeration (rapid movement) and by underwater green plants.

1 What is the name of the process by which green plants produce oxygen?

2 Where might aeration occur in a body of water?

The solubility of oxygen in water varies with temperature. Table 1 shows how much oxygen can dissolve in fresh water at various temperatures at 1 atmosphere pressure.

Temperature (°C)	1	4	8	12	16	20	24	28	32
Dissolved oxygen (ppm)	14.2	13.2	11.9	10.8	9.9	9.2	8.5	7.9	7.4

Table 1

3 Draw a 'best fit' line graph of dissolved oxygen (y-axis) against temperature (x-axis). Label the axes.

4 Describe the relationship between temperature and how much oxygen can dissolve.

The figures in table 1 relate to the maximum oxygen that can dissolve at a particular temperature. This does not mean that this quantity of oxygen will definitely be present.

5 A fish tank is 1.5 m long by 0.5 m wide by 0.5 m high. It is full to the brim with water. The tank is aerated and the water present is saturated with oxygen. Calculate the mass of oxygen dissolved in the tank at 20°C.
(Hints: 1 cm^3 of water = 1 mL; 1 ppm = 1 mg L^{-1})

Aerobic life forms are those that require oxygen for life. The amount of oxygen dissolved in water can be a very good indicator as to the health of a water ecosystem. Most natural water systems need at least 5 ppm of dissolved oxygen to support life. Because the amount of dissolved oxygen is such a good indicator of the health of a water-based ecosystem, it is important to measure the level of dissolved oxygen at regular intervals. This can be done with a titration method.

6 Define *titration*.

Another method for measuring dissolved oxygen uses an oxygen sensor connected to a data logger (see figure 1). The oxygen sensor is an electrolytic cell. The more oxygen there is in the water sample, the greater the flow of electricity. Low dissolved oxygen content can encourage organisms that use anaerobic fermentation. This process can release toxic gases into the water.

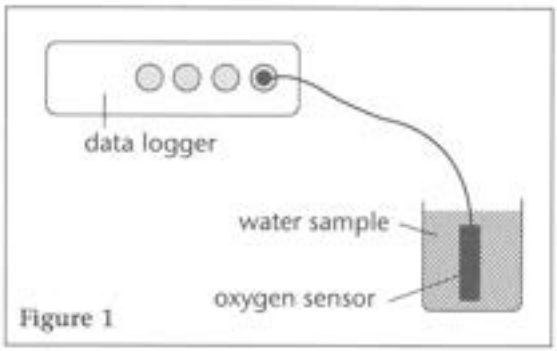


Figure 1

7 You have been asked to monitor the level of dissolved oxygen in a nearby stream over a period of time. Describe how you would do this and how you would ensure your results were valid.

8 Explain what is meant by an *anaerobic organism*.

9 A researcher found that the dissolved oxygen concentration in a body of water decreased with depth. Give a possible reason for this.

10 What is meant by *toxic*?

Research **11** What is meant by *electrolytic*?

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